```
> # DISCOVERY OF IDENTITIES FOR a_5(n)
  with(qseries) :
  T := 60 : N := 7 * T + 20 :
  f := d → etaq(q, d, N) :
  tq := q * f(7)^4 / f(1)^4 :
  p0q := q * f(14)^4 * f(1)^4 / (f(7)^4 * f(2)^4) :
  p1q := (f(14) * f(1)^7 / (f(7) * f(2)^7) − 1) / 7 :
  A5q := f(2)^4 * f(49)^5 / (q^6 * f(1)^5 * f(98)^4) :
  B5q := 1 :

> # GROUP 1, IDENTITY 1: U_A5(p0)
  m := 5 :
  S5_1_1 := series(A5q * p0q, q, 7 * T) :
  target5_1_1 := add(coeff(S5_1_1, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_1_1 := expand(findlincombo(target5_1_1, basis5, symbols5, q, 0)) :
  collect(R5_1_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((7)^4 t^3 + (3)\ (5)\ (7)^2 t^2 + (2)^3\ (7)\ t\right) p0 + \left(-(7)^3 t^2 - (2)\ (7)^2 t - (7)\right) p1 + (7)^3 t^2 + (2)\ (7)^2 t + (7) \qquad (1)$$

```
> # GROUP 1, IDENTITY 2: U_A5(p0*t^(-1))
  m := 5 :
  S5_1_2 := series(A5q * p0q / tq, q, 7 * T) :
  target5_1_2 := add(coeff(S5_1_2, q, 7 * n) * q^n, n = 0 ..T−1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_1_2 := expand(findlincombo(target5_1_2, basis5, symbols5, q, 0)) :
  collect(R5_1_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^3 t^2 - (2)^3\ (7)\ t\right) p0 + \left((7)^2 t + (7)\right) p1 - (7)^2 t - (7) \qquad (2)$$

```
> # GROUP 1, IDENTITY 3: U_A5(p0*t^(-2))
  m := 5 :
  S5_1_3 := series(A5q * p0q / tq^2, q, 7 * T) :
  target5_1_3 := series(tq * (add(coeff(S5_1_3, q, 7 * n) * q^n, n = −1 ..T−1) + O(q^T)), q,
      T) :
```

```
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ]:
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ]:
R5_1_3 := expand( findlincombo(target5_1_3, basis5, symbols5, q, 0) / t) :
collect(R5_1_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((7)^4 t^2 + (3)^2 (7)^2 t + (2)^3\right) p0 + \left(-(7)^3 t - (2)^3 (7) - \frac{1}{t}\right) p1 + (7)^3 t + (2)^3 (7) + \frac{1}{t} \quad (3)$$

```
> # GROUP 1, IDENTITY 4: U_A5(p0*t^(-3))
m := 5 :
S5_1_4 := series(A5q * p0q / tq^3, q, 7 * T) :
target5_1_4 := series(tq * (add(coeff(S5_1_4, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
     T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ]:
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ]:
R5_1_4 := expand( findlincombo(target5_1_4, basis5, symbols5, q, 0) / t) :
collect(R5_1_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((7)^7 t^4 + (7)^5 (19) t^3 + (7)^4 (11) t^2 - (3) (5) (7) (11) t - (2)^3 (11)\right) p0 + \left(-(7)^6 t^3 \right. \quad (4)$$

$$\left. - (2) (3)^2 (7)^4 t^2 - (2) (3) (7)^2 (11) t + (2) (7) (11) + \frac{(11)}{t}\right) p1 + (7)^6 t^3$$

$$+ (2) (3)^2 (7)^4 t^2 + (2) (3) (7)^2 (11) t - (2) (7) (11) - \frac{(11)}{t}$$

```
> # GROUP 1, IDENTITY 5: U_A5(p0*t^(-4))
m := 6 :
S5_1_5 := series(A5q * p0q / tq^4, q, 7 * T) :
target5_1_5 := series(tq * (add(coeff(S5_1_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
     T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ]:
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ]:
R5_1_5 := expand( findlincombo(target5_1_5, basis5, symbols5, q, 0) / t) :
collect(R5_1_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left((7)^9 t^5 + (2)^3 (7)^7 t^4 - (2) (3)^2 (5) (7)^5 t^3 - (2)^2 (5) (7)^3 (43) t^2 - (2) (5) (7)^3 t \right. \quad (5)$$

$$+ (2)^4 (3)^2 (5)) p0 + \left(-(7)^8 t^4 - (7)^7 t^3 + (2) (3)^2 (5) (7)^4 t^2 + (2) (5) (7)^3 (11) t \right.$$

$$\left. + (2) (5)^2 (7) - \frac{(2) (3)^2 (5)}{t} \right) p1 + (7)^8 t^4 + (7)^7 t^3 - (2) (3)^2 (5) (7)^4 t^2$$

$$- (2) (5) (7)^3 (11) t - (2) (5)^2 (7) + \frac{(2) (3)^2 (5)}{t}$$

> # GROUP 1, IDENTITY 6: U_A5(p0*t^(-5))
> m := 7 :
> S5_1_6 := series(A5q * p0q / tq^5, q, 7 * T) :
> target5_1_6 := series(tq * (add(coeff(S5_1_6, q, 7 * n) * q^n, n = -1 .. T - 1) + O(q^T)), q, T) :
> basis5 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q * tq^j, q, T), j = 0 .. m)] :
> symbols5 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
> R5_1_6 := expand(findlincombo(target5_1_6, basis5, symbols5, q, 0) / t) :
> collect(R5_1_6, [p0, p1, t], recursive, ifactor);

$$nx = , 24$$

$$\#\ of\ terms , 45$$

$$\left(-(7)^{11} t^6 - (2) (7)^9 (23) t^5 - (2) (3) (5) (7)^7 (19) t^4 - (7)^6 (17) (19) t^3 \right. \tag{6}$$

$$+ (3) (7)^4 (19) t^2 + (5) (7)^2 (11) (19) t - (2)^3 (3) (11) (19)) p0 + \left((7)^{10} t^5 \right.$$

$$+ (3)^2 (5) (7)^8 t^4 + (2)^2 (7)^7 (19) t^3 + (3) (5) (7)^5 (19) t^2 - (2)^2 (7)^3 (19) t$$

$$\left. - (2)^2 (7) (11) (19) + \frac{(3) (11) (19)}{t} \right) p1 - (7)^{10} t^5 - (3)^2 (5) (7)^8 t^4$$

$$- (2)^2 (7)^7 (19) t^3 - (3) (5) (7)^5 (19) t^2 + (2)^2 (7)^3 (19) t + (2)^2 (7) (11) (19)$$

$$- \frac{(3) (11) (19)}{t}$$

> # GROUP 1, IDENTITY 7: U_A5(p0*t^(-6))
> m := 8 :
> S5_1_7 := series(A5q * p0q / tq^6, q, 7 * T) :
> target5_1_7 := series(tq * (add(coeff(S5_1_7, q, 7 * n) * q^n, n = -1 .. T - 1) + O(q^T)), q, T) :
> basis5 := [seq(series(tq^j, q, T), j = 0 .. m), seq(series(p0q * tq^j, q, T), j = 0 .. m), seq(series(p1q * tq^j, q, T), j = 0 .. m)] :
> symbols5 := [seq(t^j, j = 0 .. m), seq(p0 * t^j, j = 0 .. m), seq(p1 * t^j, j = 0 .. m)] :
> R5_1_7 := expand(findlincombo(target5_1_7, basis5, symbols5, q, 0) / t) :
> collect(R5_1_7, [p0, p1, t], recursive, ifactor);

$$nx = , 27$$

of terms , 48

$$\left((7)^{13}t^7+(2)(3)^3(7)^{11}t^6+(2)(3)^3(7)^9(23)t^5+(2)(3)(5)(7)^7(19)(23)t^4\right. \tag{7}$$
$$+(7)^6(23)(389)t^3+(3)(7)^5(23)(31)t^2-(7)^2(23)(67)t$$
$$\left.+(2)^3(3)(5)(11)(23)\right)p0+\left(-(7)^{12}t^6-(7)^{10}(53)t^5-(2)^2(7)^8(13)(23)t^4\right.$$
$$-(2)^2(7)^7(19)(23)t^3-(3)^3(7)^5(13)(23)t^2-(2)(7)^3(23)(283)t$$
$$\left.+(2)(7)(19)(23)-\frac{(3)(5)(11)(23)}{t}\right)p1+(7)^{12}t^6+(7)^{10}(53)t^5$$
$$+(2)^2(7)^8(13)(23)t^4+(2)^2(7)^7(19)(23)t^3+(3)^3(7)^5(13)(23)t^2$$
$$+(2)(7)^3(23)(283)t-(2)(7)(19)(23)+\frac{(3)(5)(11)(23)}{t}$$

```
> # GROUP 2, IDENTITY 1: U_A5(p1)
  m := 6 :
  S5_2_1 := series(A5q * p1q, q, 7 * T) :
  target5_2_1 := add(coeff(S5_2_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_2_1 := expand(findlincombo(target5_2_1, basis5, symbols5, q, 0)) :
  collect(R5_2_1, [p0, p1, t], recursive, ifactor);
```

nx = , 21

of terms , 42

$$\left(-(2)^2(7)^9t^6-(2)^2(3)(7)^8t^5-(2)^2(3)^3(7)^6t^4-(2)^5(3)(5)(7)^4t^3\right. \tag{8}$$
$$\left.-(2)^3(3)(7)^2(41)t^2-(2)^3(7)(11)t\right)p0+\left((7)^{10}t^6+(2)(7)^8(11)t^5\right.$$
$$+(7)^7(29)t^4+(2)^3(7)^5(19)t^3+(2)^2(3)(7)^3(41)t^2+(2)^2(7)(13)(17)t$$
$$\left.+(7)(11)\right)p1-(5)(7)^8t^5-(3)^2(7)^6(11)t^4-(7)^4(809)t^3-(2)(7)^2(1627)t^2$$
$$-(7)(883)t-(7)(11)$$

```
> # GROUP 2, IDENTITY 2: U_A5(p1*t^(-1))
  m := 5 :
  S5_2_2 := series(A5q * p1q / tq, q, 7 * T) :
  target5_2_2 := add(coeff(S5_2_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_2_2 := expand(findlincombo(target5_2_2, basis5, symbols5, q, 0)) :
  collect(R5_2_2, [p0, p1, t], recursive, ifactor);
```

nx = , 18

of terms , 39

$$(7)\,t \tag{9}$$

```
> # GROUP 2, IDENTITY 3: U_A5(p1*t^(-2))
  m := 5 :
  S5_2_3 := series(A5q * p1q / tq^2, q, 7 * T) :
  target5_2_3 := series(tq * (add(coeff(S5_2_3, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R5_2_3 := expand( findlincombo(target5_2_3, basis5, symbols5, q, 0) / t) :
  collect(R5_2_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^3\,t^2-(2)^6\,(7)\,t-(2)^3\right)p0+\left((2)^3\,(7)^2\,t+(3)\,(19)+\frac{1}{t}\right)p1-(7)^3\,t^2 \tag{10}$$

$$-(2)\,(7)\,(29)\,t-(3)\,(19)-\frac{1}{t}$$

```
> # GROUP 2, IDENTITY 4: U_A5(p1*t^(-3))
  m := 5 :
  S5_2_4 := series(A5q * p1q / tq^3, q, 7 * T) :
  target5_2_4 := series(tq * (add(coeff(S5_2_4, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R5_2_4 := expand( findlincombo(target5_2_4, basis5, symbols5, q, 0) / t) :
  collect(R5_2_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^2\,(7)^5\,t^3+(2)^2\,(7)^3\,(23)\,t^2+(2)^2\,(3)^3\,(5)\,(7)\,t+(2)^4\,(5)\right)p0+\left(-(2)^2\,(7)^4\,t^2 \tag{11}$$

$$-(7)^2\,(89)\,t-(2)\,(241)-\frac{(2)\,(5)}{t}\right)p1+(7)^5\,t^3+(2)\,(7)^3\,(19)\,t^2$$

$$+(3)\,(7)\,(211)\,t+(2)\,(241)+\frac{(2)\,(5)}{t}$$

```
> # GROUP 2, IDENTITY 5: U_A5(p1*t^(-4))
  m := 5 :
  S5_2_5 := series(A5q * p1q / tq^4, q, 7 * T) :
  target5_2_5 := series(tq * (add(coeff(S5_2_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
```

```
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R5_2_5 := expand(findlincombo(target5_2_5, basis5, symbols5, q, 0) / t) :
collect(R5_2_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^2\,(5)\,(7)^5\,t^3-(2)^2\,(7)^3\,(11)^2\,t^2-(2)^2\,(7)^2\,(109)\,t-(2)^3\,(79)\right)p0+\left((3)\,(7)^5\,t^2+(2)^3\,(7)^2\,(59)\,t+(2)^3\,(7)^3+\frac{(79)}{t}\right)p1-(7)^7\,t^4-(2)\,(3)^2\,(7)^5\,t^3-(3)\,(7)^3\,(79)\,t^2-(2)^5\,(3)\,(5)\,(7)^2\,t-(2)^3\,(7)^3-\frac{(79)}{t} \tag{12}$$

```
> # GROUP 2, IDENTITY 6: U_A5(p1*t^(-5))
  m := 6 :
  S5_2_6 := series(A5q * p1q / tq^5, q, 7 * T) :
  target5_2_6 := series(tq * (add(coeff(S5_2_6, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_2_6 := expand(findlincombo(target5_2_6, basis5, symbols5, q, 0) / t) :
  collect(R5_2_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left((2)^3\,(7)^7\,t^4+(2)^2\,(5)\,(7)^6\,t^3+(2)^2\,(7)^4\,(73)\,t^2+(2)^2\,(7)^2\,(491)\,t+(2)^3\,(5)\,(109)\right)p0+\left(-(7)^7\,t^3-(2)^2\,(5)\,(7)^5\,t^2-(2)\,(7)^3\,(11)\,(13)\,t-(2)\,(5)\,(7)\,(179)-\frac{(5)\,(109)}{t}\right)p1-(7)^9\,t^5+(5)^2\,(7)^6\,t^3+(2)\,(5)\,(7)^4\,(23)\,t^2+(2)\,(3)\,(7)^2\,(11)\,(31)\,t+(2)\,(5)\,(7)\,(179)+\frac{(5)\,(109)}{t} \tag{13}$$

```
> # GROUP 2, IDENTITY 7: U_A5(p1*t^(-6))
  m := 7 :
  S5_2_7 := series(A5q * p1q / tq^6, q, 7 * T) :
  target5_2_7 := series(tq * (add(coeff(S5_2_7, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
```

```
R5_2_7 := expand(findlincombo(target5_2_7, basis5, symbols5, q, 0)/t):
collect(R5_2_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\begin{aligned}&\left(-(2)^2\,(7)^{11}\,t^6-(2)^2\,(7)^9\,(29)\,t^5-(2)^2\,(5)\,(7)^7\,(73)\,t^4-(2)^4\,(7)^7\,(11)\,t^3\right.\\&\quad\left.-(2)^3\,(5)\,(7)^4\,(89)\,t^2-(2)^3\,(3)^3\,(7)^2\,(37)\,t-(2)^{11}\,(13)\right)p0+\left((2)^2\,(7)^{10}\,t^5\right.\\&\quad+(2)^4\,(7)^9\,t^4+(2)\,(7)^7\,(97)\,t^3+(7)^5\,(1123)\,t^2+(2)\,(7)^3\,(1613)\,t\\&\quad\left.+(3)^3\,(5)^2\,(7)\,(11)+\frac{(2)^8\,(13)}{t}\right)p1-(7)^9\,(11)\,t^5-(2)^3\,(5)\,(7)^7\,(17)\,t^4\\&\quad-(7)^6\,(1433)\,t^3-(3)\,(7)^4\,(2797)\,t^2-(3)\,(7)^2\,(7603)\,t-(2)\,(13)\,(1999)\\&\quad-\frac{(2)^8\,(13)}{t}\end{aligned}\tag{14}$$

```
> # GROUP 3, IDENTITY 1: U_A5(1)
m := 6:
S5_3_1 := series(A5q, q, 7*T):
target5_3_1 := add(coeff(S5_3_1, q, 7*n)*q^n, n=0..T-1)+O(q^T):
basis5 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)]:
symbols5 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)]:
R5_3_1 := expand(findlincombo(target5_3_1, basis5, symbols5, q, 0)):
collect(R5_3_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&\left((2)^2\,(7)^{10}\,t^6+(2)^2\,(7)^8\,(23)\,t^5+(2)^2\,(3)^2\,(7)^6\,(23)\,t^4+(2)^5\,(3)\,(7)^4\,(37)\,t^3\right.\\&\quad\left.+(2)^5\,(7)^3\,(31)\,t^2+(2)^3\,(7)\,(79)\,t\right)p0+\left(-(2)^2\,(7)^9\,t^5-(7)^7\,(89)\,t^4\right.\\&\quad-(2)^3\,(7)^5\,(97)\,t^3-(3)^2\,(7)^3\,(359)\,t^2-(2)^4\,(5)\,(7)^2\,(11)\,t-(7)\,(79)\left.\right)p1\\&\quad+(7)^{10}\,t^6+(3)^2\,(5)\,(7)^8\,t^5+(7)^6\,(727)\,t^4+(7)^4\,(11)^2\,(47)\,t^3+(7)^2\,(22807)\,t^2\\&\quad+(7)\,(61)\,(101)\,t+(7)\,(79)\end{aligned}\tag{15}$$

```
> # GROUP 3, IDENTITY 2: U_A5(t^(-1))
m := 5:
S5_3_2 := series(A5q/tq, q, 7*T):
target5_3_2 := series(tq*(add(coeff(S5_3_2, q, 7*n)*q^n, n=-1..T-1)+O(q^T)), q,
    T):
basis5 := [seq(series(tq^j, q, T), j=0..m), seq(series(p0q*tq^j, q, T), j=0..m), seq(series(p1q
    *tq^j, q, T), j=0..m)]:
symbols5 := [seq(t^j, j=0..m), seq(p0*t^j, j=0..m), seq(p1*t^j, j=0..m)]:
```

```
R5_3_2 := expand( findlincombo(target5_3_2, basis5, symbols5, q, 0) / t) :
collect(R5_3_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left((2)^3\ (7)\ t + (2)^3 \right) p0 + \left(-(2)^3 - \frac{1}{t} \right) p1 + (7)\ t + (2)^3 + \frac{1}{t} \tag{16}$$

```
> # GROUP 3, IDENTITY 3: U_A5(t^(-2))
  m := 5 :
  S5_3_3 := series(A5q / tq^2, q, 7 * T) :
  target5_3_3 := series(tq * (add(coeff(S5_3_3, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R5_3_3 := expand( findlincombo(target5_3_3, basis5, symbols5, q, 0) / t) :
  collect(R5_3_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left(-(2)^3\ (7)\ t - (2)^3\ (3) \right) p0 + \left((2)\ (5) + \frac{(3)}{t} \right) p1 - (7)^3\ t^2 - (2)\ (7)\ t - (2)\ (5) \tag{17}$$

$$- \frac{(3)}{t}$$

```
> # GROUP 3, IDENTITY 4: U_A5(t^(-3))
  m := 5 :
  S5_3_4 := series(A5q / tq^3, q, 7 * T) :
  target5_3_4 := series(tq * (add(coeff(S5_3_4, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m) ] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
  R5_3_4 := expand( findlincombo(target5_3_4, basis5, symbols5, q, 0) / t) :
  collect(R5_3_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms , 39$$

$$\left((2)^2\ (7)^5\ t^3 + (2)^2\ (7)^4\ t^2 - (2)^2\ (7)^2\ t + (2)^4\ (5) \right) p0 + \left(-(2)^2\ (7)^4\ t^2 - (2)^3\ (3)\ (7)^2\ t \right. \tag{18}$$

$$\left. + (2)\ (7) - \frac{(2)\ (5)}{t} \right) p1 + (7)^5\ t^3 + (2)\ (7)^3\ (19)\ t^2 + (2)\ (7)\ (89)\ t - (2)\ (7)$$

$$+ \frac{(2)\ (5)}{t}$$

```
> # GROUP 3, IDENTITY 5: U_A5(t^(-4))
  m := 5 :
  S5_3_5 := series(A5q / tq^4, q, 7 * T) :
  target5_3_5 := series(tq * (add(coeff(S5_3_5, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_3_5 := expand(findlincombo(target5_3_5, basis5, symbols5, q, 0) / t) :
  collect(R5_3_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^7\,t^4-(2)^2\,(7)^5\,(43)\,t^3-(2)^2\,(5)\,(7)^5\,t^2-(2)^2\,(3)\,(7)^3\,t-(2)^3\,(11)\right)p0 + \left((2)^3\,(7)^6\,t^3+(2)^2\,(7)^4\,(41)\,t^2+(2)^2\,(7)^3\,(31)\,t+(2)^2\,(7)\,(19)+\frac{(11)}{t}\right)p1 - (7)^7\,t^4-(2)\,(7)^5\,(37)\,t^3-(2)\,(7)^3\,(619)\,t^2-(2)^2\,(3)\,(7)^2\,(73)\,t-(2)^2\,(7)\,(19) - \frac{(11)}{t} \tag{19}$$

```
> # GROUP 3, IDENTITY 6: U_A5(t^(-5))
  m := 6 :
  S5_3_6 := series(A5q / tq^5, q, 7 * T) :
  target5_3_6 := series(tq * (add(coeff(S5_3_6, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T)), q,
      T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_3_6 := expand(findlincombo(target5_3_6, basis5, symbols5, q, 0) / t) :
  collect(R5_3_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left(-(2)^3\,(7)^9\,t^5-(2)^4\,(3)\,(7)^7\,t^4+(2)^2\,(5)\,(7)^7\,t^3+(2)^2\,(7)^4\,(13)\,(23)\,t^2+(2)^2\,(3)^2\,(7)^2\,(29)\,t-(2)^3\,(11)\,(23)\right)p0+\left((2)^3\,(7)^8\,t^4+(2)\,(3)\,(7)^7\,t^3-(2)^3\,(7)^5\,(17)\,t^2-(2)\,(3)^2\,(7)^3\,(59)\,t-(2)\,(7)\,(443)+\frac{(11)\,(23)}{t}\right)p1-(7)^9\,t^5-(2)^3\,(7)^8\,t^4-(2)^3\,(3)\,(7)^6\,t^3+(2)\,(7)^4\,(521)\,t^2+(2)\,(7)^2\,(3739)\,t+(2)\,(7)\,(443)-\frac{(11)\,(23)}{t} \tag{20}$$

```
> # GROUP 3, IDENTITY 7: U_A5(t^(-6))
```

```
m := 7 :
S5_3_7 := series(A5q / tq^6, q, 7 * T) :
target5_3_7 := series(tq * (add(coeff(S5_3_7, q, 7 * n) * q^n, n = -1 ..T-1) + O(q^T) ), q,
    T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R5_3_7 := expand( findlincombo(target5_3_7, basis5, symbols5, q, 0) / t) :
collect(R5_3_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\begin{aligned}&\left((2)^2\,(7)^{11}\,t^6 + (2)^2\,(5)\,(7)^9\,(13)\,t^5 + (2)^2\,(7)^8\,(113)\,t^4 + (2)^5\,(7)^6\,(41)\,t^3\right.\\&\quad\left. - (2)^3\,(7)^4\,(11)\,(37)\,t^2 - (2)^3\,(7)^2\,(11)\,(53)\,t + (2)^4\,(7)\,(13)\,(19)\right)p0 + \Bigg(\\&\quad -(2)^2\,(7)^{10}\,t^5 - (7)^8\,(257)\,t^4 - (2)\,(3)\,(7)^7\,(71)\,t^3 - (7)^5\,(13)\,(89)\,t^2\\&\quad + (2)\,(7)^3\,(11)\,(131)\,t + (5)\,(7)\,(17)\,(43) - \frac{(2)\,(7)\,(13)\,(19)}{t}\Bigg)p1\\&\quad + (3)^2\,(5)\,(7)^9\,t^5 + (7)^7\,(11)\,(173)\,t^4 + (3)^2\,(7)^6\,(17)\,(19)\,t^3 + (3)^2\,(7)^4\,(29)^2\,t^2\\&\quad - (7)^2\,(23)\,(887)\,t - (2)^4\,(3)\,(13)\,(41) + \frac{(2)\,(7)\,(13)\,(19)}{t}\end{aligned} \tag{21}$$

```
> # GROUP 4, IDENTITY 1: U_B5(p0)
m := 5 :
S5_4_1 := series(B5q * p0q, q, 7 * T) :
target5_4_1 := add(coeff(S5_4_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R5_4_1 := expand( findlincombo(target5_4_1, basis5, symbols5, q, 0) ) :
collect(R5_4_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\begin{aligned}&\left((2)^3\,(7)^7\,t^4 + (2)^4\,(5)\,(7)^5\,t^3 + (2)^3\,(3)^3\,(7)^3\,t^2 + (7)\,(79)\,t\right)p0 + \left((7)^{10}\,t^5\right.\\&\quad\left. + (7)^8\,(19)\,t^4 + (2)\,(3)^2\,(7)^7\,t^3 + (3)\,(7)^4\,(109)\,t^2 + (2)\,(7)^3\,(17)\,t + (2)^2\right)p1\\&\quad + (7)^9\,t^5 + (7)^7\,(11)\,t^4 + (2)\,(7)^5\,(19)\,t^3 + (7)^3\,(31)\,t^2 + (2)\,(3)\,(7)\,t\end{aligned} \tag{22}$$

```
> # GROUP 4, IDENTITY 2: U_B5(p0*t^(-1))
m := 5 :
S5_4_2 := series(B5q * p0q / tq, q, 7 * T) :
target5_4_2 := add(coeff(S5_4_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
```

```
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m)] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
R5_4_2 := expand(findlincombo(target5_4_2, basis5, symbols5, q, 0)) :
collect(R5_4_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^3\, t^2 - (2)^3\, (7)\, t\right) p0 + \left((7)^2\, t + (7)\right) p1 + 1 - (7)^2\, t \tag{23}$$

```
> # GROUP 4, IDENTITY 3: U_B5(p0*t^(-2))
  m := 5 :
  S5_4_3 := series(B5q * p0q / tq^2, q, 7 * T) :
  target5_4_3 := add(coeff(S5_4_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_4_3 := expand(findlincombo(target5_4_3, basis5, symbols5, q, 0)) :
  collect(R5_4_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^5\, t^3 - (2)^3\, (7)^3\, t^2\right) p0 + \left((7)^4\, t^2 + (7)^3\, t\right) p1 - (7)^4\, t^2 - (7)^3\, t - (2)^2\, (3) \tag{24}$$

```
> # GROUP 4, IDENTITY 4: U_B5(p0*t^(-3))
  m := 5 :
  S5_4_4 := series(B5q * p0q / tq^3, q, 7 * T) :
  target5_4_4 := add(coeff(S5_4_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
      * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_4_4 := expand(findlincombo(target5_4_4, basis5, symbols5, q, 0)) :
  collect(R5_4_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(1 + (2)\,(3)\,(7)^7\, t^4 + (2)\,(7)^5\,(31)\, t^3 + (3)^3\,(7)^4\, t^2 + (2)\,(5)\,(7)^2\, t\right) p0 + \left(\right. \tag{25}$$
$$\left. -(2)\,(3)\,(7)^6\, t^3 - (2)^3\,(7)^5\, t^2 - (3)\,(7)^4\, t - (2)^2\,(7)\right) p1 + (2)\,(7)^6\, t^3$$
$$+ (2)^3\,(3)\,(7)^4\, t^2 + (7)^2\,(83)\, t + (2)^2\,(3)^3$$

```
> # GROUP 4, IDENTITY 5: U_B5(p0*t^(-4))
  m := 5 :
  S5_4_5 := series(B5q * p0q / tq^4, q, 7 * T) :
  target5_4_5 := add(coeff(S5_4_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
```

```
    * tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R5_4_5 := expand( findlincombo(target5_4_5, basis5, symbols5, q, 0) ) :
collect(R5_4_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(7)^9 t^5 - (2)(5)^2 (7)^7 t^4 - (2)^2 (3)(5)(7)^6 t^3 - (2)(3)^4 (7)^4 t^2 - (2)^2 (3)(5)(7)^2 t - (2)(3)\right) p0 + \left((7)^8 t^4 + (7)^8 t^3 + (2)(3)^3 (7)^5 t^2 + (2)(3)^2 (7)^4 t + (2)^3 (3)(7)\right) p1 - (7)^8 t^4 - (5)^2 (7)^6 t^3 - (2)(3)(7)^4 (31) t^2 - (2)(3)(7)^2 (83) t - (2)^3 (3)(5)(7) \quad (26)$$

```
> # GROUP 4, IDENTITY 6: U_B5(p0*t^(-5))
m := 6 :
S5_4_6 := series(B5q * p0q / tq^5, q, 7 * T) :
target5_4_6 := add(coeff(S5_4_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R5_4_6 := expand( findlincombo(target5_4_6, basis5, symbols5, q, 0) ) :
collect(R5_4_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\left(-(7)^{11} t^6 - (2)^3 (7)^9 t^5 + (3)(7)^9 t^4 + (2)(3)(5)(7)^7 t^3 + (3)^4 (7)^5 t^2 + (2)(3)(5)(7)^3 t + (3)(7)\right) p0 + \left((7)^{10} t^5 + (7)^9 t^4 - (3)(7)^8 t^3 - (3)^3 (7)^6 t^2 - (3)^2 (7)^5 t - (2)^2 (3)(7)^2\right) p1 - (7)^{10} t^5 - (7)^9 t^4 + (3)^2 (7)^7 t^3 + (3)(7)^5 (31) t^2 + (3)(7)^3 (83) t + (2)^2 (7)(11)(19) \quad (27)$$

```
> # GROUP 4, IDENTITY 7: U_B5(p0*t^(-6))
m := 7 :
S5_4_7 := series(B5q * p0q / tq^6, q, 7 * T) :
target5_4_7 := add(coeff(S5_4_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R5_4_7 := expand( findlincombo(target5_4_7, basis5, symbols5, q, 0) ) :
collect(R5_4_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 24$$

$$\#\ of\ terms\ , 45$$

$$\left(-(7)^{13} t^7 - (2)^3 (7)^{11} t^6 + (2)(7)^9 t^4 + (2)^2 (5)(7)^7 t^3 + (2)(3)^3 (7)^5 t^2 + (2)^2 (5)(7)^3 t + (2)(7)\right) p0 + \left((7)^{12} t^6 + (7)^{11} t^5 - (2)(7)^8 t^3 - (2)(3)^2 (7)^6 t^2\right. \quad (28)$$

$$-(2)\,(3)\,(7)^5\,t-(2)^3\,(7)^2)\,p1-(7)^{12}\,t^6-(7)^{11}\,t^5+(2)\,(3)\,(7)^7\,t^3$$
$$+(2)\,(7)^5\,(31)\,t^2+(2)\,(7)^3\,(83)\,t-(2)^2\,(7)^2\,(11)\,(17)$$

```
> # GROUP 5, IDENTITY 1: U_B5(p1*t^(-1))
  m := 5 :
  S5_5_1 := series(B5q * p1q / tq, q, 7 * T) :
  target5_5_1 := add(coeff(S5_5_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_5_1 := expand(findlincombo(target5_5_1, basis5, symbols5, q, 0)) :
  collect(R5_5_1, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^3\,(7)^8\,t^4+(2)^4\,(5)\,(7)^6\,t^3+(2)^3\,(3)^3\,(7)^4\,t^2+(2)^3\,(3)\,(7)\,(23)\,t\right)p0+\left((7)^{11}\,t^5\right. \tag{29}$$
$$\left.+(7)^9\,(19)\,t^4+(2)\,(3)^2\,(7)^8\,t^3+(3)\,(7)^5\,(109)\,t^2+(2)\,(7)^4\,(17)\,t+(29)\right)p1-1$$
$$+(7)^{10}\,t^5+(7)^8\,(11)\,t^4+(2)\,(7)^6\,(19)\,t^3+(7)^4\,(31)\,t^2+(7)\,(41)\,t$$

```
> # GROUP 5, IDENTITY 2: U_B5(p1*t^(-2))
  m := 5 :
  S5_5_2 := series(B5q * p1q / tq^2, q, 7 * T) :
  target5_5_2 := add(coeff(S5_5_2, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_5_2 := expand(findlincombo(target5_5_2, basis5, symbols5, q, 0)) :
  collect(R5_5_2, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^3\,t^2-(2)^3\,(7)^2\,t\right)p0+\left((2)^3\,(7)^2\,t+(7)^2\right)p1-(7)^3\,t^2-(2)^3\,(7)^2\,t+(11) \tag{30}$$

```
> # GROUP 5, IDENTITY 3: U_B5(p1*t^(-3))
  m := 5 :
  S5_5_3 := series(B5q * p1q / tq^3, q, 7 * T) :
  target5_5_3 := add(coeff(S5_5_3, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
       * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_5_3 := expand(findlincombo(target5_5_3, basis5, symbols5, q, 0)) :
  collect(R5_5_3, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^5\,t^3-(2)^3\,(7)^4\,t^2\right)p0+\left((2)^3\,(7)^4\,t^2+(7)^4\,t\right)p1-(7)^5\,t^3-(2)^3\,(7)^4\,t^2-(7)^4\,t-(2)^5\,(3) \tag{31}$$

```
> # GROUP 5, IDENTITY 4: U_B5(p1*t^(-4))
  m := 5 :
  S5_5_4 := series(B5q * p1q / tq^4, q, 7 * T) :
  target5_5_4 := add(coeff(S5_5_4, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_5_4 := expand(findlincombo(target5_5_4, basis5, symbols5, q, 0)) :
  collect(R5_5_4, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left((2)^4\,(3)\,(7)^7\,t^4+(2)^3\,(3)^2\,(7)^6\,t^3+(2)^3\,(3)^3\,(7)^4\,t^2+(2)^4\,(5)\,(7)^2\,t+(2)^3\right)p0+\left(-(2)^4\,(3)\,(7)^6\,t^3-(5)\,(7)^5\,(13)\,t^2-(2)^3\,(3)\,(7)^4\,t-(2)^5\,(7)\right)p1-(7)^7\,t^4+(2)^4\,(7)^6\,t^3+(7)^4\,(199)\,t^2+(2)^3\,(7)^2\,(83)\,t+(2)^2\,(5)\,(37) \tag{32}$$

```
> # GROUP 5, IDENTITY 5: U_B5(p1*t^(-5))
  m := 5 :
  S5_5_5 := series(B5q * p1q / tq^5, q, 7 * T) :
  target5_5_5 := add(coeff(S5_5_5, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
  R5_5_5 := expand(findlincombo(target5_5_5, basis5, symbols5, q, 0)) :
  collect(R5_5_5, [p0, p1, t], recursive, ifactor);
```

$$nx = , 18$$

$$\#\ of\ terms\ , 39$$

$$\left(-(2)^3\,(7)^9\,t^5-(2)^6\,(7)^8\,t^4-(2)^4\,(5)\,(7)^7\,t^3-(2)^3\,(3)^3\,(7)^5\,t^2-(2)^4\,(5)\,(7)^3\,t-(2)^3\,(7)\right)p0+\left((2)^3\,(7)^8\,t^4+(3)^2\,(7)^8\,t^3+(2)^3\,(3)^2\,(7)^6\,t^2+(2)^3\,(3)\,(7)^5\,t+(2)^5\,(7)^2\right)p1-(7)^9\,t^5-(2)^3\,(7)^8\,t^4-(7)^7\,(31)\,t^3-(2)^3\,(7)^5\,(31)\,t^2-(2)^3\,(7)^3\,(83)\,t-(2)^2\,(5)\,(7)\,(37) \tag{33}$$

```
> # GROUP 5, IDENTITY 6: U_B5(p1*t^(-6))
  m := 6 :
  S5_5_6 := series(B5q * p1q / tq^6, q, 7 * T) :
  target5_5_6 := add(coeff(S5_5_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
        * tq^j, q, T), j = 0 ..m)] :
  symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m)] :
```

```
R5_5_6 := expand( findlincombo(target5_5_6, basis5, symbols5, q, 0) ) :
collect(R5_5_6, [p0, p1, t], recursive, ifactor);
```

$$nx = , 21$$

$$\#\ of\ terms\ , 42$$

$$\begin{aligned}&\left(-(2)^3\,(7)^{11}\,t^6-(2)^3\,(7)^{10}\,t^5+(2)^3\,(5)\,(7)^9\,t^4+(2)^4\,(5)^2\,(7)^7\,t^3+(2)^3\,(3)^3\,(5)\,(7)^5\,t^2\right.\\&\quad\left.+(2)^4\,(5)^2\,(7)^3\,t+(2)^3\,(5)\,(7)\right)p0+\left((2)^3\,(7)^{10}\,t^5+(7)^{10}\,t^4-(2)^3\,(5)\,(7)^8\,t^3\right.\\&\quad\left.-(2)^3\,(3)^2\,(5)\,(7)^6\,t^2-(2)^3\,(3)\,(5)\,(7)^5\,t-(2)^5\,(5)\,(7)^2\right)p1-(7)^{11}\,t^6\\&\quad-(2)^3\,(7)^{10}\,t^5-(7)^{10}\,t^4+(2)^3\,(3)\,(5)\,(7)^7\,t^3+(2)^3\,(5)\,(7)^5\,(31)\,t^2\\&\quad+(2)^3\,(5)\,(7)^3\,(83)\,t+(2)^4\,(5)\,(7)\,(59)\end{aligned}\tag{34}$$

```
> # GROUP 5, IDENTITY 7: U_B5(p1*t^(-7))
m := 8 :
S5_5_7 := series(B5q * p1q / tq^7, q, 7 * T) :
target5_5_7 := series(tq * (add(coeff(S5_5_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) ), q, T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m), seq(series(p0q * tq^j, q, T), j = 0 ..m), seq(series(p1q
    * tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m), seq(p0 * t^j, j = 0 ..m), seq(p1 * t^j, j = 0 ..m) ] :
R5_5_7 := expand( findlincombo(target5_5_7, basis5, symbols5, q, 0) / t) :
collect(R5_5_7, [p0, p1, t], recursive, ifactor);
```

$$nx = , 27$$

$$\#\ of\ terms\ , 48$$

$$\begin{aligned}&\left(-(2)^3\,(7)^{13}\,t^7-(2)^3\,(7)^{12}\,t^6-(2)^3\,(7)^9\,(17)\,t^4-(2)^4\,(5)\,(7)^7\,(17)\,t^3\right.\\&\quad\left.-(2)^3\,(3)^3\,(7)^5\,(17)\,t^2-(2)^4\,(5)\,(7)^3\,(17)\,t-(2)^3\,(7)\,(17)\right)p0+\left((2)^3\,(7)^{12}\,t^6\right.\\&\quad\left.+(2)\,(3)\,(7)^{11}\,t^5-(7)^9\,(19)\,t^4+(2)\,(7)^7\,(409)\,t^3+(3)\,(7)^6\,(389)\,t^2\right.\\&\quad\left.+(2)\,(7)^4\,(1399)\,t+(7)^4\,(11)+\frac{1}{t}\right)p1+(2)\,(3)\,(7)^{13}\,t^7+(2)^2\,(5)\,(7)^{12}\,t^6\\&\quad+(2)^4\,(7)^{10}\,(17)\,t^5+(3)\,(7)^8\,(631)\,t^4+(3)\,(5)\,(7)^7\,(29)\,t^3-(7)^5\,(2953)\,t^2\\&\quad-(2)^2\,(3)^2\,(5)\,(7)^3\,(59)\,t-(7)\,(151)\,(179)\end{aligned}\tag{35}$$

```
> # GROUP 6, IDENTITY 1: U_B5(1)
m := 1 :
S5_6_1 := series(B5q, q, 7 * T) :
target5_6_1 := add(coeff(S5_6_1, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m) ] :
symbols5 := [seq(t^j, j = 0 ..m) ] :
R5_6_1 := expand( findlincombo(target5_6_1, basis5, symbols5, q, 0) ) :
collect(R5_6_1, [t], recursive, ifactor);
```

$$nx = , 2$$

```
# of terms , 23
1
```
(36)

```
> # GROUP 6, IDENTITY 2: U_B5(t^(-1))
  m := 1 :
  S5_6_2 := series(B5q/tq, q, 7*T) :
  target5_6_2 := add(coeff(S5_6_2, q, 7*n)*q^n, n=0..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j=0..m)] :
  symbols5 := [seq(t^j, j=0..m)] :
  R5_6_2 := expand(findlincombo(target5_6_2, basis5, symbols5, q, 0)) :
  collect(R5_6_2, [t], recursive, ifactor);
```
```
nx = , 2
# of terms , 23
-(7) t - (2)^2
```
(37)

```
> # GROUP 6, IDENTITY 3: U_B5(t^(-2))
  m := 2 :
  S5_6_3 := series(B5q/tq^2, q, 7*T) :
  target5_6_3 := add(coeff(S5_6_3, q, 7*n)*q^n, n=0..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j=0..m)] :
  symbols5 := [seq(t^j, j=0..m)] :
  R5_6_3 := expand(findlincombo(target5_6_3, basis5, symbols5, q, 0)) :
  collect(R5_6_3, [t], recursive, ifactor);
```
```
nx = , 3
# of terms , 24
-(7)^3 t^2 + (2)^2 (5)
```
(38)

```
> # GROUP 6, IDENTITY 4: U_B5(t^(-3))
  m := 3 :
  S5_6_4 := series(B5q/tq^3, q, 7*T) :
  target5_6_4 := add(coeff(S5_6_4, q, 7*n)*q^n, n=0..T-1) + O(q^T) :
  basis5 := [seq(series(tq^j, q, T), j=0..m)] :
  symbols5 := [seq(t^j, j=0..m)] :
  R5_6_4 := expand(findlincombo(target5_6_4, basis5, symbols5, q, 0)) :
  collect(R5_6_4, [t], recursive, ifactor);
```
```
nx = , 4
# of terms , 25
-(7)^5 t^3 - (2)^3 (11)
```
(39)

```
> # GROUP 6, IDENTITY 5: U_B5(t^(-4))
  m := 4 :
  S5_6_5 := series(B5q/tq^4, q, 7*T) :
  target5_6_5 := add(coeff(S5_6_5, q, 7*n)*q^n, n=0..T-1) + O(q^T) :
```

```
basis5 := [seq(series(tq^j, q, T), j = 0 ..m)] :
symbols5 := [seq(t^j, j = 0 ..m)] :
R5_6_5 := expand(findlincombo(target5_6_5, basis5, symbols5, q, 0)) :
collect(R5_6_5, [t], recursive, ifactor);
```

$$nx = , 5$$

$$\#\ of\ terms\ , 26$$

$$-(7)^{7}\, t^{4} + (2)^{2}\, (5)\, (13) \tag{40}$$

```
> # GROUP 6, IDENTITY 6: U_B5(t^(-5))
m := 5 :
S5_6_6 := series(B5q / tq^5, q, 7 * T) :
target5_6_6 := add(coeff(S5_6_6, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m)] :
symbols5 := [seq(t^j, j = 0 ..m)] :
R5_6_6 := expand(findlincombo(target5_6_6, basis5, symbols5, q, 0)) :
collect(R5_6_6, [t], recursive, ifactor);
```

$$nx = , 6$$

$$\#\ of\ terms\ , 27$$

$$-(7)^{9}\, t^{5} + (2)^{2}\, (7)\, (17) \tag{41}$$

```
> # GROUP 6, IDENTITY 7: U_B5(t^(-6))
m := 6 :
S5_6_7 := series(B5q / tq^6, q, 7 * T) :
target5_6_7 := add(coeff(S5_6_7, q, 7 * n) * q^n, n = 0 ..T-1) + O(q^T) :
basis5 := [seq(series(tq^j, q, T), j = 0 ..m)] :
symbols5 := [seq(t^j, j = 0 ..m)] :
R5_6_7 := expand(findlincombo(target5_6_7, basis5, symbols5, q, 0)) :
collect(R5_6_7, [t], recursive, ifactor);
```

$$nx = , 7$$

$$\#\ of\ terms\ , 28$$

$$-(7)^{11}\, t^{6} - (2)^{3}\, (7)\, (13)\, (23) \tag{42}$$

```
>
```